PIPING AND INSTRUMENTATION DIAGRAM STP SYSTEM

PROJECT :

PT BATUTUA TEMBAGA RAYA

PT. GRINVIRO BIOTEKNO INDONESIA

Water and Waste Water Treatment System

2026

DOCUMENT REVISION RECORD

DATE	REVISION	DESCRIPTION
16/07/2026	00	- P&ID STP 50 CMD First Issued
27/07/2026	01	Piping and valve specification revision, addition of level sensor at the inlet tank, pump voltage specification revision, blower configuration revised to non-VFD operation, PSV revision, and drawing annotation update.

STATUS : DETAIL ENGINEERING DESIGN

PROJECT
STP 50 CMD

OWNER
PT BATUTUA TEMBAGA RAYA

DIKETAHUI OLEH/ACKNOWLEDGED BY:

PROJECT LEADER

KONTRAKTOR UTAMA / MAIN CONTRACTOR

PT. GRINVIRO BIOTEKNO INDONESIA
Water and Waste Water Treatment System

PROJECT MANAGER

DISETUJUI OLEH/APPROVE BY:

PT. GRINVIRO BIOTEKNO INDONESIA

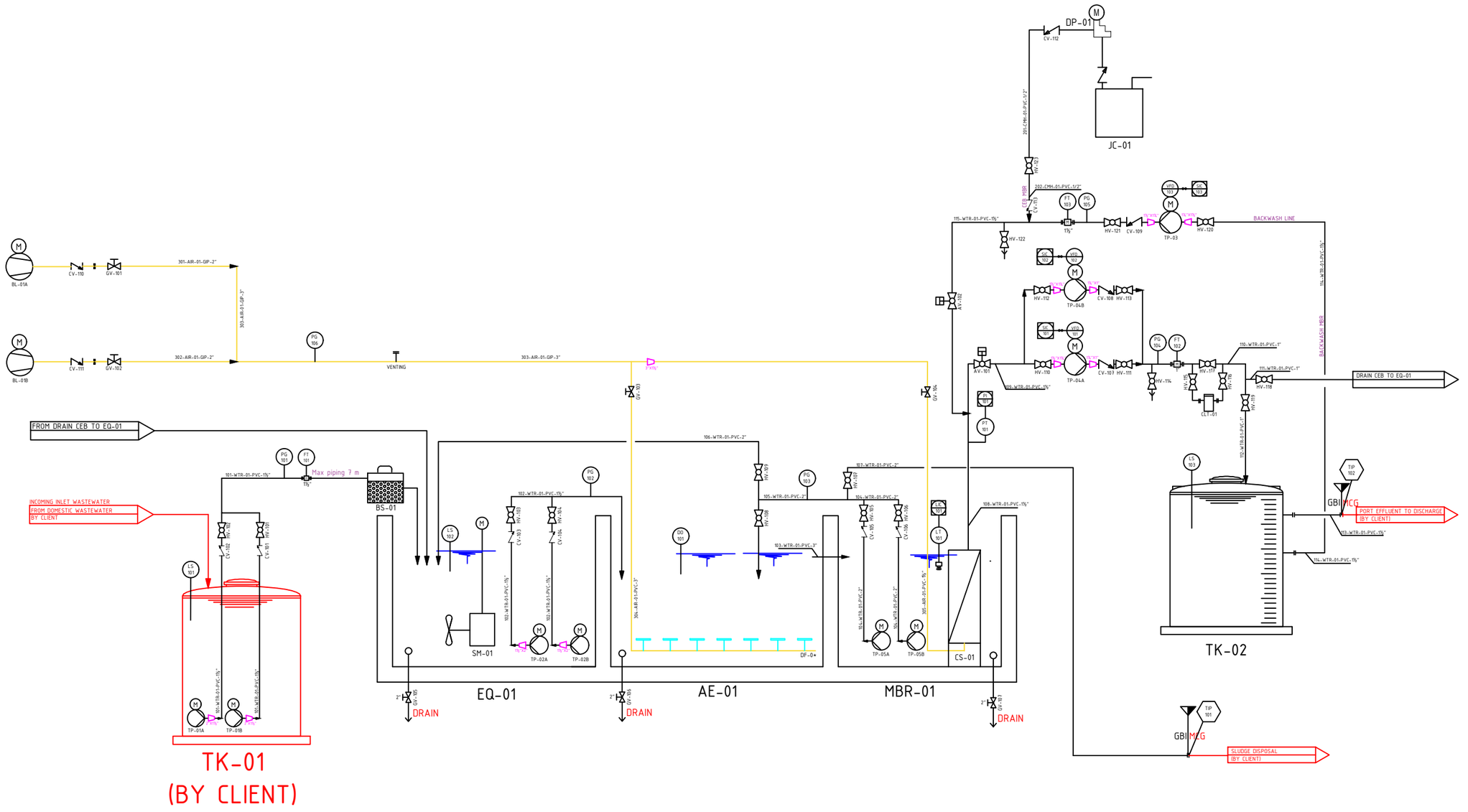
AGUNG WIBOWO
PROJECT MANAGER

NAMA GAMBAR/ DRAWING NAME :

PIPING AND INSTRUMENT DIAGRAM

SKALA / SCALE	PARAF/ SIGN	TANGGAL/ DATE
NO SCALE		
DIGAMBAR OLEH/DRAWN. BY: PAT		27-07-2026
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TAG.NO	TP-01A/B
EQUIPMENT	INLET TRANSFER PUMP
TYPE.	SUBMERSIBLE PUMP
CAPACITY	2.5 CMH HEAD 11 m
POWER	0.75 kW/3 PHASE/380 V
QTY.	2 UNITS (1W+1S)

TAG.NO	TP-02A/B
EQUIPMENT	EQ TRANSFER PUMP
TYPE.	SUBMERSIBLE PUMP
CAPACITY	2.5 CMH HEAD 7-10 m
POWER	0.75 kW/3 PHASE/380 V
QTY.	2 UNITS (1W+1S)

TAG.NO	TP-03
EQUIPMENT	MBR BACKWASH PUMP
TYPE.	VERTICAL MULTISTAGE PUMP
CAPACITY	6 CMH HEAD 26.8 m
POWER	0.75 kW/3 PHASE/380 V
QTY.	1 UNIT

TAG.NO	TP-04A/B
EQUIPMENT	MBR PERMEATE PUMP
TYPE.	VERTICAL MULTISTAGE PUMP
CAPACITY	3 CMH HEAD 21 m
POWER	0.55 kW/3 PHASE/380 V
QTY.	2 UNITS (1W+1S)

TAG.NO	BL-01A/B
EQUIPMENT	MBR BLOWER
TYPE	SIDE CHANNEL BLOWER
CAPACITY	2.75-2.95 Nm3/min
POWER	3 kW/3 PHASE/200-240V/345-415Y
QTY.	2 UNITS (1W+1S)

TAG.NO	TP-05A/B
EQUIPMENT	MBR RAS PUMP
TYPE	SUBMERSIBLE PUMP
CAPACITY	10 CMH HEAD 10 m
POWER	0.75 kW//3 PHASE/380 V
QTY.	2 UNITS (1W+1S)

TAG.NO	EQ-01
EQUIPMENT	EQUALIZING TANK
VOLUME	18 m^3
MATERIAL	CARBON STEEL
QTY.	1 UNIT
SYSTEM	BUILD IN PACKAGE

TAG.NO	AE-01
EQUIPMENT	AERATION TANK
VOLUME	29.6 m^3
MATERIAL	CARBON STEEL
QTY.	1 UNIT
SYSTEM	BUILD IN PACKAGE

TAG.NO	MBR-01
EQUIPMENT	MEMBRANE BIO REACTOR TANK
VOLUME	1 SKID
MATERIAL	CARBON STEEL
QTY.	1 UNIT
SYSTEM	BUILD IN PACKAGE

TAG.NO	TK-01
EQUIPMENT	INLET TANK
VOLUME	BY CLIENT
MATERIAL	-
QTY.	1 UNIT

TAG.NO	CLT-01
EQUIPMENT	CHLORINATION
QTY	1 UNIT

TAG.NO	CS-01
EQUIPMENT	CASSETE MEMBRANE
TYPE	PVDF

TAG.NO	JC-01
EQUIPMENT	CHLORINE JERRY CAN
CAPACITY	50 L
QTY.	1 UNIT

TAG.NO	TK-02
EQUIPMENT	BACKWASH TANK
VOLUME	3000 L
MATERIAL	PE
QTY.	1 UNIT

TAG.NO	BS-01
EQUIPMENT	BASKET SCREEN
QTY.	1 UNIT

TAG.NO	DF-0*
EQUIPMENT	DIFFUSER AERATION
QTY.	TBD

TAG.NO	DP-01
EQUIPMENT	CHLORINE DOSING PUMP
QTY.	1 UNIT

TAG.NO	SM-01
EQUIPMENT	SUBMERSIBLE MIXER
QTY.	1 UNIT @ 0.75 kW

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LEGENDS

VALVE SYMBOLS

- BUTTERFLY VALVE
- FLEXIBLE JOINT
- CHECK VALVE
- BALL VALVE
- AUTOMATIC VALVE
- BUTTERFLY SELENOID VALVE
- FOOT VALVE
- PRESSURE SAVETY VALVE
- BREATH VALVE
- LOOSE JOINT
- AUTOMATIC EXHAUST VALVE

ABBREVIATIONS INSTRUMENT & CONTROLS

AE	ANALYZER ELEMENT
AIT	ANALYZER TRANSMITTER
AIC	ANALUZER INDICATOR CONTROL
FIT	FLOW INDICATOR TRANSMITTER
FQI	FLOW QUANTITY INDICATOR
H/O/A	HAND OFF AUTO SWITCH
HS	HAND SWITCH
LG	LEVEL GAUGE
LT	LEVEL TRANSMITTER
LIC	LEVEL INDICATOR CONTROL
LS	LEVEL SWITCH
MEI	MAN EQUIPMENT INDICATOR
PG	PRESSURE GAUGE
PT	PRESSURE TRANSMITTER
PIC	PRESSURE INDICATOR CONTROL
SIC	SPEED INDICATOR CONTROL
VFD	VARIABLE FREQUENCY DRIVE

BALLON INSTRUMENTS

- FIELD MOUNTED
- NORMALLY ACCESSIBLE TO AN OPERATOR

- FIELD OR LOCAL CONTROL PANEL
- NORMALLY ACCESSIBLE TO AN OPERATOR

- CENTRAL OR MAIN CONTROL ROOM
- NOT NORMALLY ACCESSIBLE TO AN OPERATOR

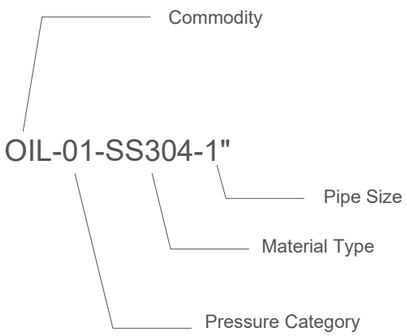
- BASIC PROCESS CONTROL SYSTEM (BPCS)
- VISIBLE IN FIELD OR FIELDS MOUNTED

- BASIC PROCESS CONTROL SYSTEM (BPCS)
- DISPLAY IN CENTRAL OR MAIN CONTROL ROOM

EQUIPMENT SYMBOLS

- ELECTROMAGNETIC FLOWMETER
- GENERAL BLOWER
- GENERAL PUMP
- DOSING PUMP
- ROTARY DRUM SCREEN
- SUBMERSIBLE MIXER
- VERTICAL MIXER AGITATOR

PIPE TAGGING



NOTE FOR PIPE TAGGING:

1. Line Number
   - 10* = Main Interconnection Pipe
   - 20* = Chemical Line Pipe
   - 30* = Air/Biogas Line Pipe
   - * Line Pipe Sequences

2. Commodity
   - WTR = Water Reject
   - SLG = Sludge
   - BGS = Biogas
   - AIR = Air
   - OIL= Oil
   - CLW = WATER

3. Pipe Pressure Category
   - 01 = 0-5 bar
   - 02 = >5-10 bar
   - 03 = >10-15 bar

4. Material Type
   - PVC = PVC Pipe
   - SS304 = SS304 Pipe
   - PVC = PVC Pipe
   - SS316L = SS316L Pipe

5. Pipe Size
   - 1" , 11/2", 2", 21/2", 3" dst

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